JUAN C. SANCHEZ-ARIAS, MD, PHD

Data Scientist • Physician-Scientist • HealthTech & AI Systems Leader

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PROFESSIONAL PROFILE

Physician-Scientist, Data Scientist, and Google-Certified Generative AI Leader with over 10 years of combined clinical, biomedical research, and healthtech industry experience. Specialized in engineering production-grade Generative AI pipelines, RAG architectures, and biomedical knowledge graphs that accelerate preclinical drug discovery and solve complex biological data problems. Proven track record leading cross-functional teams of PhD scientists and ML engineers, bridging clinical insights with modern data engineering to build compliant, scalable, and impactful AI products.

CORE COMPETENCIES & TECHNICAL SKILLS

AI & LLM Systems: Generative AI, Prompt Engineering, RAG Frameworks, LLM Evaluation & Benchmarking, Knowledge Graphs, Biomedical NLP.

Programming & Data: Python (pandas, NumPy, scikit-learn, PyTorch/HuggingFace), R (tidyverse, Bioconductor), SQL (PostgreSQL, BigQuery), Bash, Git.

Life Sciences Domain: Preclinical Drug Discovery, Clinical Medicine, Target Validation, Bio-Ontologies, Live-Cell Bio-Image Processing (ImageJ/Fiji), Computational Biology.

Product & Leadership: Product Ownership (CSPO), Cross-Functional Team Leadership, Agile/Scrum, Stakeholder Management, Scientific Due Diligence, GCP, Docker.

WORK EXPERIENCE

Data Scientist — BenchSci
Jun 2026 – Present | Remote (Toronto, ON)

Enterprise AI platform accelerating preclinical drug discovery for leading global pharmaceutical organizations.

- Architect production-grade Generative AI, RAG, and agentic workflows to extract, synthesize, and contextualize high-dimensional preclinical and therapeutics data for top global pharmaceutical clients.
- Develop automated LLM benchmarking and evaluation pipelines to ensure high scientific accuracy, hallucination mitigation, and strict domain compliance across complex oncology and neuroscience indications.
- Scale biomedical knowledge graph coverage and improve semantic vector search retrieval across millions of research papers, patents, and customer internal experimental records.

Science Tech Lead & Team Lead — BenchSci
Jun 2023 – May 2026 | Remote (Toronto, ON)

- Managed and mentored a cross-functional team of PhD scientists and data specialists responsible for ingesting, validating, and structuring proprietary enterprise pharmaceutical datasets.
- Served as **Product Owner (CSPO)** for the Customer Internal Data E2E processing pipeline, dramatically accelerating ingestion velocity and client time-to-value.
- Led the R&D and deployment of Generative AI applications to extract, summarize, and structure unstructured experimental records in Electronic Lab Notebooks (ELNs) and IP documents.
- Acted as subject matter expert for Neuroscience and Oncology therapeutic areas, shaping platform roadmap with executive stakeholders and technical product managers.

Scientist II — BenchSci
Jun 2022 – May 2023 | Remote (Toronto, ON)

- Engineered automated data ingestion and ontology mapping pipelines in Python/SQL for preclinical assays and biological reagents, reducing manual review cycles by over 30%.
- Partnered directly with pharmaceutical partners to map bespoke scientific schemas to standardized knowledge graph ontologies.

WORK EXPERIENCE (CONTINUED)

Postdoctoral Research Fellow (Health Research BC Fellow) — **University of Victoria** Nov 2020 – Apr 2022 | Victoria, BC

Division of Medical Sciences

- Recipient of the **Michael Smith Health Research BC Trainee Award** investigating cardiac arrhythmia and neurodevelopmental channelopathies (*ANK2* p.S646F) in BC Indigenous communities.
- Bridged preclinical genetics, live-cell fluorescence microscopy (calcium & voltage dynamics), and computational modeling with patient ECG data to provide clinical diagnostic insights.

Graduate Research Assistant (Ph.D. Candidate) — **University of Victoria** Sep 2015 – Aug 2020 | Victoria, BC

- Built custom Python and ImageJ bio-image analysis pipelines for automated quantification of 3D dendritic spine morphology and synaptic plasticity in cortical neurons.
- Taught neuroanatomy and histology laboratory curricula to 1st/2nd year medical students (Island Medical Program, UBC/UVic).

Pre-Diploma Medical Intern & Clinical Researcher — **Hospital Universitario del Valle** 2013 – 2014 | Cali, Colombia

- Completed full-time clinical rotations across Internal Medicine, General Surgery, Pediatrics, OB/GYN, and Emergency Care in a tertiary-care university hospital.
- Conducted clinical research and bedside patient evaluations with the Department of Neurosurgery, contributing to clinical protocols and neurocritical care management.

EDUCATION

Doctor of Philosophy (Ph.D.) in Neuroscience — University of Victoria, BC, Canada 2015 – 2020

Doctor of Medicine (MD / Médico y Cirujano) — Universidad del Valle, Cali, Colombia 2007 – 2014

CERTIFICATIONS & PROFESSIONAL TRAINING

Google Certified Generative AI Leader (Google Cloud, 2025) • **Google Data Analytics Professional Specialization** (Google Cloud, 2026) • **Certified Scrum Product Owner (CSPO - 1920762)** (Scrum Alliance, 2024) • **Genetic Epidemiology & Epigenetics** (Univ. del Valle, 2022) • **Code in Place (Python Engineering)** (Stanford Univ., 2021).

SELECTED AWARDS & HONORS

Research Trainee Award — *Michael Smith Health Research BC* 2021 – 2022

Nb Gulila Star Award — *International Gap Junction Conference* 2019

John & Myrtle Tilley Graduate Scholarship — *University of Victoria* 2019

James & Laurette Agnew Memorial Scholarship — *University of Victoria* 2015 – 2019

School of Medicine Yearly Dean's List — *Universidad del Valle* 2007, 2009, 2013, 2014